

Syed Shayan Mazahir

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EDUCATION

University of Waterloo

Bachelor of Geospatial Data Science, Minor in Computer Science

Waterloo, ON

Expected June 2030

PROJECTS

ATOM — Humanoid Robot, Phase 0 Complete | C++, Python, ESP32, OpenCV/MediaPipe, UART, PWM, ADC

- Built a real-time gesture-mirroring robotic hand end to end: 21-landmark hand tracking → dot-product joint angles → per-user open/fist calibration → 5 calibrated 0–180° servo positions over UART at 20 Hz
- Authored ESP32 firmware (C++) and a config-driven Python pipeline (shared JSON protocol for servo map, port, and tuning) with rolling-average smoothing, rate limiting, and range clamping on both ends of the serial link for fault-tolerant servo control
- Debugged a non-functional EMG acquisition chain from first principles: derived the expected ADC baseline from the sensor's 1.5 V reference, identified 20 Hz undersampling as an aliasing artifact, and isolated laptop USB/mains coupling as the dominant noise source through controlled single-variable testing (96% → 15% rail clipping)
- Found and documented two defects in the vendor's published example code (16-bit `int` overflow in `sq()` on AVR; integer-division truncation of the signal envelope) while maintaining an 800-line controlled-experiment log with null controls and retracted results
- Designed the power architecture for a 5V/3A rail with a 2.5A servo load budget and isolated logic and actuator supplies; organized 3D-print files by verified support requirements and published a web-based STL browser via GitHub Pages

Mr. Beep-A-Lot — Multi-Mode Autonomous Robot | C++, Arduino, Processing (Java), I2C, Serial

- Built an autonomous robot with three operational modes: obstacle avoidance, environment scanning, and human following
- Integrated IR remote control, dual ultrasonic sensors, a servo-controlled radar, and an I2C LCD display, with sensor fusion driving autonomous navigation
- Developed a Processing (Java) visualization for real-time radar data over Serial communication

games.random — AI-Powered Game Development Platform | JavaScript, Claude API, Monaco Editor

- Engineered few-shot prompt templates for the Claude API, achieving 85%+ syntactically correct code generation from natural language descriptions
- Built a real-time validation pipeline with the Monaco editor, reducing runtime errors by 40% via instant syntax highlighting and error feedback
- Open-sourced on GitHub with documentation of the LLM prompt engineering techniques used

EXPERIENCE

Robotics Team Lead & Programming Hub Mentor

2024 – 2025

Louise Arbour Secondary School

Brampton, ON

- Led a 15-student team to First Place in the RobOlympics competition
- Designed and built 4 autonomous robots, including line-following and soccer robots
- Implemented A* pathfinding and PID control algorithms in C++; integrated ultrasonic/IR sensors, gyroscopes, and motor encoders for real-time positioning
- Engineered voltage regulation circuits (5V to 15V), electrical schematics, and wiring harnesses; specified and documented custom mechanical assemblies
- Built a Flask web application with Google API integration serving 500+ students, including OAuth 2.0 authentication and SSL/HTTPS
- Mentored 20+ students in Python and data structures, trained 15 in embedded C++ and circuit debugging with oscilloscopes and multimeters, and maintained annotated schematics and Git repositories

TECHNICAL SKILLS

Languages: C/C++, Python, JavaScript, Java, Racket

Hardware & Embedded: Arduino, ESP32, Raspberry Pi, VEX, UART/Serial, I²C, PWM, ADC, Ultrasonic/IR/Gyro/EMG Sensors, Servo & Motor Control, PID, Circuit Design, Oscilloscope & Multimeter

Tools & Libraries: OpenCV/MediaPipe, Digital Filtering & Signal Acquisition, 3D Printing (FDM), Git/GitHub, Node.js, Flask, REST APIs